

CS 636 Semester 2024–2025-II: Assignment 1

20th January 2025

Due Your assignment is due by Feb 9, 2025, 11:59 PM IST.

General Policies

- You should do this assignment ALONE.
- Do not copy or turn in solutions from other sources. You will be PENALIZED if caught.

Submission

- Submission will be through Canvas.
- Submit a compressed file called “`<roll>-assign1.tar.gz`”. The compressed file should have the following structure.

```
-- roll-assign1
-- -- roll-assign1.pdf
-- -- <roll>
-- -- -- <source-files>
```

Use the PDF file to include details (e.g., compilation command, optimizations, challenges, and results) about your solution. Name the `.cpp` and `.h` source files using your roll number.

- We encourage you to use the L^AT_EX typesetting system for generating the PDF file. You can use tools like Tikz, Inkscape, or Draw.io for drawing figures if required. You can alternatively upload a scanned copy of a handwritten solution, but MAKE SURE the submission is legible.
- You will get up to TWO LATE days to submit your assignment, with a 25% penalty for each day.

Evaluation

- Write your programs such that the EXACT output format (if any) is respected.
- We will evaluate your implementations on a Unix-like system, for example, recent Debian-based distributions installed on KD first floor labs.
- We will evaluate the implementations with our inputs and test cases, so remember to test thoroughly.

Problem

[100 marks]

Implement the DJIT⁺ and FastTrack algorithms as discussed in class. Here are the references to the two papers for more details.

DJIT⁺

- E. Pozniarsky and A. Schuster. Efficient On-the-Fly Data Race Detection in Multi-threaded C++ Programs. PPOPP, 2003.
- E. Pozniarsky and A. Schuster. MultiRace: efficient on-the-fly data race detection in multithreaded C++ programs. CCPE, vol. 19, no. 3, pp. 327–340, Mar. 2007.

FastTrack

- C. Flanagan and S. Freund. FastTrack: Efficient and Precise Dynamic Data Race Detection. PLDI, 2009.
- Cormac Flanagan and Stephen N. Freund. 2017. The FastTrack2 Race Detector. Technical Report. Williams College.

Use Piazza to discuss questions related to the papers.

We will test your implementation using traces of concurrent program runs. The traces will include thread fork/join, memory read and write, and lock acquire and release logs. Use [this sample trace](#) for your implementation. The traces are regular ASCII files. Here are a few arbitrary entries from the sample trace.

```
Thread begin: 0
TID: 0, IP: 0x74538c822211, ADDR: 0x74538c83ae06, Size (B): 1, isRead: 1
TID: 0, IP: 0x74538c83aab0, ADDR: 0x74538c82222f, Size (B): 8, isRead: 0
...
Before pthread_create(): Parent: 0
...
TID: 1, IP: 0x745378a60d88, ADDR: 0x745375fffc0, Size (B): 8, isRead: 1
Thread begin: 2
...
TID: 2, IP: 0x74538c80f1ab, ADDR: 0x74538c839a28, Size (B): 8, isRead: 1
Before lock release: TID: 2, Lock address: 0x74538c83aa08
TID: 2, IP: 0x745378aa1a64, ADDR: 0x74538c83aa18, Size (B): 4, isRead: 1
...
TID: 1, IP: 0x745375fffc08, ADDR: 0x74538c828058, Size (B): 8, isRead: 0
After lock release: TID: 2, Lock address: 0x74538c83aa08
TID: 2, IP: 0x7453753ffe08, ADDR: 0x745378b840d5, Size (B): 8, isRead: 0
...
TID: 2, IP: 0x7453753ffba0, ADDR: 0x745378df3585, Size (B): 8, isRead: 0
Before lock acquire: TID: 1, Lock address: 0x74538c83aa08
TID: 1, IP: 0x745378a9ffe4, ADDR: 0x74538c83aa18, Size (B): 4, isRead: 1
...
TID: 1, IP: 0x74538c83aa14, ADDR: 0x745378aa0051, Size (B): 4, isRead: 0
After lock acquire: TID: 1, Lock address: 0x74538c83aa08
```

```
TID: 2, IP: 0x7453753ffba8, ADDR: 0x745378df3585, Size (B): 8, isRead: 0
...
Thread ended: 2
```

IP stands for instruction pointer. The rest of the content should be readily interpretable. The traces have been generated with Intel Pin. You need not worry about errors while parsing the trace files.

Implementation Your implementation should accept the following arguments:

algo Either `djit` or `fasttrack`

trace Absolute path to the input trace file

Output

- (i) The output should list all unique data races that you found in the trace for both the algorithms. Given a trace, both DJIT⁺ and FastTrack should have the same coverage.

For each data race, report the address, the type of race (W-W, W-R, or R-W), and the threads involved (TID:1 TID: 3). A data race can manifest several times during the same execution (i.e., same address, same type, and same threads). Therefore, print the count of occurrences for each unique race. A race report on address location **X** involving threads **A** and **B** is the same as the race reported on location **X** involving **B** and **A**. A output line should be formatted as follows.

```
0x745378df3585 W-W TID:1 TID:3 Count: 56
```

Leave the output blank if there are no races with the input trace.

- (ii) Compare the performance of the two algorithms (in seconds). Disable the print statements for the performance runs. Report the average of a few runs for more reliable performance estimates.